Second-Order Linear Homogeneous Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The auxiliary equation**

Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.

$r^2 - 5r + 6 = 0$.

2 **[Foundational]** • **Target: The auxiliary equation**

Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.

$2r^2 + 4r - 6 = 0$, which simplifies to $r^2 + 2r - 3 = 0$.

3 **[Foundational]** • **Target: The auxiliary equation**

Explain why substituting $y = e^{rx}$ into $ay'' + by' + cy = 0$ produces the auxiliary equation $ar^2 + br + c = 0$.

$y = e^{rx}$ gives $y' = re^{rx}$ and $y'' = r^2e^{rx}$. Substituting: $ar^2e^{rx} + bre^{rx} + ce^{rx} = 0$, i.e., $e^{rx}(ar^2 + br + c) = 0$. Since $e^{rx} \neq 0$ for all x, this forces $ar^2 + br + c = 0$.

4 **[Foundational]** • **Target: Distinct real roots**

Solve $y'' - 9y = 0$ (note this factors as a difference of squares in r).

$r^2 - 9 = (r - 3)(r + 3) = 0 \Rightarrow r = \pm 3$. General solution: $y = C_1e^{3x} + C_2e^{-3x}$.

5 **[Foundational]** • **Target: Repeated roots**

Explain why, for a repeated root r, $y_2 = xe^{rx}$ (not just a second copy of e^{rx}) is needed to build the general solution.

$y_1 = e^{rx}$ and a constant multiple of it, ce^{rx}, are linearly dependent — they don't give two independent solutions needed for a second-order equation's two-dimensional solution space. Direct substitution confirms $y_2 = xe^{rx}$ also solves the ODE when r is a repeated root, and y_1, y_2 are linearly independent (their Wronskian is $e^{2rx} \neq 0$), so together they span the full solution space.

6 **[Foundational]** • **Target: Classifying without fully solving**

Without solving, state whether $y'' + 6y' + 5y = 0$ has real, repeated, or complex roots, using the discriminant $b^2 - 4ac$ of the auxiliary equation.

Discriminant $= 36 - 20 = 16 > 0$ and a perfect square, so the roots are real and distinct ($r = -1, -5$).

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Classifying without fully solving

Without solving, classify the roots of $y'' + 10y' + 25y = 0$.

Discriminant $= 100 - 100 = 0$, so the auxiliary equation has one repeated real root ($r = -5$).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Application: a damped spring

A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).

$r^2 + 4r + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, so $r = -2 \pm i$. General solution: $y = e^{-2x}(C_1 \cos x + C_2 \sin x)$ — this is under-damped (complex roots with negative real part give oscillatory decay).

9 [Intermediate] • Target: Application: a damped spring

Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

$r^2 + 6r + 9 = (r+3)^2 = 0 \Rightarrow r = -3$ (repeated) — critically damped. So $y = (C_1 + C_2 x)e^{-3x}$. From $y(0) = 1$: $C_1 = 1$. $y' = C_2 e^{-3x} - 3(C_1 + C_2 x)e^{-3x}$; $y'(0) = C_2 - 3C_1 = 0 \Rightarrow C_2 = 3$. Thus $y = (1 + 3x)e^{-3x}$.

10 [Intermediate] • Target: Application: a damped spring

For the general damped spring equation $my'' + cy' + ky = 0$ ($m, c, k > 0$), explain in one sentence why the real part of every root of the auxiliary equation is always negative (so every solution decays).

By the quadratic formula the roots are $r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$: when the roots are real, $\sqrt{c^2 - 4mk} < c$ (since $mk > 0$), so both roots stay negative after adding $-c$ and dividing by $2m > 0$; when complex, the real part is exactly $-\frac{c}{2m} < 0$ since $c, m > 0$ — either way the exponential factor $e^{-(c/2m)x}$ (or the two negative real exponentials) forces every solution toward 0 as $x \to \infty$.

11 [Intermediate] • Target: Verifying general solutions and the superposition principle

Verify by direct substitution that $y = C_1 e^{2x} + C_2 e^{3x}$ solves $y'' - 5y' + 6y = 0$ for any constants C_1, C_2.

$y' = 2C_1 e^{2x} + 3C_2 e^{3x}$, $y'' = 4C_1 e^{2x} + 9C_2 e^{3x}$. Then $y'' - 5y' + 6y = (4 - 10 + 6)C_1 e^{2x} + (9 - 15 + 6)C_2 e^{3x} = 0 + 0 = 0$ for any C_1, C_2.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Verifying general solutions and the superposition principle

State the superposition principle for a linear homogeneous ODE, and explain why it guarantees that $y = C_1y_1 + C_2y_2$ solves $ay'' + by' + cy = 0$ whenever y_1, y_2 each do.

If y_1, y_2 both solve $ay'' + by' + cy = 0$, then so does any linear combination $C_1y_1 + C_2y_2$, because the operator $y \to ay'' + by' + cy$ is linear: $a(C_1y_1 + C_2y_2)'' + b(\cdots)' + c(\cdots) = C_1(ay_1'' + by_1' + cy_1) + C_2(ay_2'' + by_2' + cy_2) = C_1(0) + C_2(0) = 0$.

13 [Intermediate] • Target: Verifying general solutions and the superposition principle

Find the auxiliary equation's roots for $y'' - 4y = 0$, then verify that both $y = e^{2x}$ and $y = e^{-2x}$ individually satisfy the ODE.

$r^2 - 4 = 0 \Rightarrow r = \pm 2$. For $y = e^{2x}$: $y'' = 4e^{2x}$, so $y'' - 4y = 4e^{2x} - 4e^{2x} = 0$. For $y = e^{-2x}$: $y'' = 4e^{-2x}$, so $y'' - 4y = 4e^{-2x} - 4e^{-2x} = 0$. Both check out.

14 [Intermediate] • Target: Boundary value problems

Solve the boundary value problem $y'' + y = 0$, $y(0) = 0$, $y(\pi/2) = 3$.

$r^2 + 1 = 0 \Rightarrow r = \pm i$, so $y = C_1\cos x + C_2\sin x$. $y(0) = C_1 = 0$. $y(\pi/2) = C_2\sin(\pi/2) = C_2 = 3$. Thus $y = 3\sin x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Boundary value problems • Cross-Lesson: 1.01 Separable Equations

Explain why the boundary value problem $y'' + 4y = 0$, $y(0) = 0$, $y(\pi) = 1$ has NO solution, while $y(0) = 0$, $y(\pi/4) = 1$ does.

General solution: $y = C_1\cos 2x + C_2\sin 2x$. $y(0) = C_1 = 0$, so $y = C_2\sin 2x$. At $x = \pi$: $\sin 2\pi = 0$, so $y(\pi) = 0$ for every C_2 — the condition $y(\pi) = 1$ can never be met. At $x = \pi/4$: $\sin(\pi/2) = 1 \neq 0$, so $C_2 = 1$ solves $y(\pi/4) = 1$ uniquely.

16 [Challenging] • Target: Boundary value problems • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Find the indicial equation and roots about $x = 0$ for the differential equation $2x^2y'' - xy' + (x - 5)y = 0$, and state the form of the two linearly independent Frobenius solutions.

Step 1: Check regular singularity at $x_0 = 0$: $P(x) = -\frac{1}{2x}$, $\quad Q(x) = \frac{x-5}{2x^2}$.
$p_0 = \lim_{x\to 0} xP(x) = -\frac{1}{2}$, $\quad q_0 = \lim_{x\to 0} x^2Q(x) = -\frac{5}{2}$. Both limits are finite, confirming $x = 0$ is a regular singular point. Step 2: Indicial equation: $r(r-1) + p_0r + q_0 = r^2 - r - \frac{1}{2}r - \frac{5}{2} = 0 \Rightarrow 2r^2 - 3r - 5 = 0$. Factoring: $(2r - 5)(r + 1) = 0 \Rightarrow r_1 = \frac{5}{2}$, $\quad r_2 = -1$. Step 3: Root difference.

© ClipSAT — clipsat.org | All Rights Reserved

17 **[Challenging]** • **Target: The auxiliary equation** • **Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

Find the general solution of the differential equation $y'' + y = 4\sin x + 2e^{-x}$.

Step 1: Complementary function: $r^2 + 1 = 0 \Rightarrow r = \pm i \Rightarrow y_c(x) = C_1\cos x + C_2\sin x$. Step 2: Particular solution trial form: For $4\sin x$: roots $\pm i$ match y_c (multiplicity $s = 1$), so trial is $y_{p1} = x(A\cos x + B\sin x)$. For $2e^{-x}$: $r = -1$ is not a root of $r^2 + 1 = 0$, so trial is $y_{p2} = Ce^{-x}$. Total trial: $y_p = x(A\cos x + B\sin x) + Ce^{-x}$. Step 3: Solve for y_{p1}.

18 **[Challenging]** • **Target: Distinct real roots** • **Cross-Lesson: 5.01 Variation of Parameters**

Find the general real solution of $\mathbf{x}' = (-2,\ -5;\ 1,\ 0)\mathbf{x}$, classify the origin, and determine whether trajectories rotate clockwise or counterclockwise.

Step 1: Characteristic polynomial: $\det(-2-\lambda,\ -5;\ 1,\ -\lambda) = \lambda(\lambda+2) + 5 = \lambda^2 + 2\lambda + 5 = 0 \Rightarrow \lambda = -1 \pm 2i$. Since $\alpha = -1 < 0$, the origin is an asymptotically stable spiral sink. Step 2: Eigenvector for $\lambda = -1 + 2i$: $(-1-2i,\ -5;\ 1,\ 1-2i)(v_1; v_2) = \mathbf{0} \Rightarrow v_1 + (1-2i)v_2 = 0 \Rightarrow \mathbf{v} = (-1+2i;\ 1) = (-1;\ 1) + i(2;\ 0)$. Step 3: Real and i.

19 **[Challenging]** • **Target: Repeated roots** • **Cross-Lesson: 1.01 Separable Equations**

[Systems of Linear Differential Equations, Phase Plane Topology & Matrix Exponentials · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 **[Challenging]** • **Target: Complex roots** • **Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Solve the double-resonance equation $y'' - 6y' + 9y = 4e^{3t}$ with $y(0) = 1$ and $y'(0) = 3$.

Step 1: Homogeneous: $r^2 - 6r + 9 = (r-3)^2 = 0 \Rightarrow y_c(t) = (C_1 + C_2 t)e^{3t}$. Step 2: Resonance multiplication: Both e^{3t} and te^{3t} are in y_c (multiplicity $s = 2$). Multiply trial by t^2: $y_p(t) = At^2e^{3t}$. Using operator $(D-3)^2[At^2e^{3t}] = e^{3t}D^2[At^2] = e^{3t}(2A) = 4e^{3t} \Rightarrow 2A = 4 \Rightarrow A = 2$. Thus $y_p(t) = 2t^2e^{3t}$. Step 3: Initial conditions: $y(t) = (C_1 + C_2 t + 2t^2)e^{3t}$. $y(0) = C_1 = 1$..

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: The auxiliary equation

Consider the second-order linear homogeneous differential equation $y'' - 2y' + ky = 0$. We are given that $y_1(x) = e^{3x}$ is a solution to this equation. \\ (a) Find the value of the parameter k. \\ (b) Find the general solution to the differential equation.

a. $k = -3, y(x) = c_1e^{3x} + c_2e^{-x}$ ✓

b. $k = 3, y(x) = c_1e^{3x} + c_2xe^{3x}$

c. $k = -3, y(x) = c_1e^{3x} + c_2e^{x}$

d. $k = 9, y(x) = c_1e^{3x} + c_2e^{-3x}$

Part (a): \\ If $y_1(x) = e^{3x}$ is a solution, then its derivatives must satisfy the equation.\\ $y_1'(x) = 3e^{3x}$\\ $y_1''(x) = 9e^{3x}$\\ Substitute these into the differential equation $y'' - 2y' + ky = 0$:\\ $9e^{3x} - 2(3e^{3x}) + k(e^{3x}) = 0$\\ $9e^{3x} - 6e^{3x} + ke^{3x} = 0$\\ Factor out e^{3x} (which is never zero):\\ $e^{3x}(9 - 6 + k) = 0$\\ $3 + k = 0$\\ So, $k = -3$.\\ \\ Part (b): \\ Now that we know $k = -3$, the differential equation is $y'' - 2y' - 3y = 0$.\\ The auxiliary (characteristic).

22 [Foundational] • Target: Distinct real roots

Consider the second-order linear homogeneous differential equation $y'' - 3y' + 2y = 0$. Find the general solution to this equation.

a. $y(x) = c_1e^{x} + c_2e^{2x}$ ✓

b. $y(x) = c_1e^{-x} + c_2e^{-2x}$

c. $y(x) = c_1\cos(x) + c_2\sin(x)$

d. $y(x) = c_1xe^{x} + c_2e^{x}$

The characteristic equation for the differential equation $y'' - 3y' + 2y = 0$ is $r^2 - 3r + 2 = 0$. Factoring this quadratic equation gives $(r - 1)(r - 2) = 0$. The roots are $r_1 = 1$ and $r_2 = 2$. Since the roots are real and distinct, the general solution is of the form $y(x) = c_1e^{r_1x} + c_2e^{r_2x}$. Substituting the roots, we get $y(x) = c_1e^{x} + c_2e^{2x}$. Choice B is incorrect because it uses negative roots. Choice C is incorrect because it corresponds to complex roots. Choice D is incorrect becau.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Repeated roots

Consider the second-order linear homogeneous differential equation $y'' + P(x)y' + Q(x)y = 0$. For the equation $x^2y'' + xy' + (x^2 - u^2)y = 0$ (Bessel's equation of order u), the point $x = 0$ is a regular singular point. If we assume a series solution of the form $y(x) = \sum_{n=0}^{\infty} a_n x^{n+r}$, where $a_0 eq 0$, the indicial equation is given by $r^2 - u^2 = 0$. For this problem, let $u = 1$, so the indicial equation is $r^2 - 1 = 0$. This yields two distinct roots, $r_1 = 1$ and $r_2 = -1$. For the larger root $r_1 = 1$, the power series solution takes the form $y_1(x) = x + \sum_{n=1}^{\infty} a_n x^{n+1}$. Substituting this into Bessel's equation of order 1, and after considerable algebraic manipulation, the recurrence relation for the coefficients a_n is found to be $a_n = -\frac{a_{n-2}}{n(n+2)}$ for $n \geq 2$. Given $a_0 = 1$, find the value of a_3 for the solution corresponding to the larger root $r_1 = 1$.

a. **0** ✓

b. 1/24

c. 1/48

d. 1/12

The recurrence relation is $a_n = -\frac{a_{n-2}}{n(n+2)}$ for $n \geq 2$, with $a_0 = 1$. We need to find a_3. For $n = 2$, $a_2 = -\frac{a_0}{2(2+2)} = -\frac{1}{2(4)} = -\frac{1}{8}$. For $n = 3$, $a_3 = -\frac{a_{3-2}}{3(3+2)} = -\frac{a_1}{3(5)} = -\frac{a_1}{15}$. Since the recurrence relation only relates terms that are two indices apart, and we are given a_0, any odd-indexed coefficients will depend on a_1. However, the form of the solution for Bessel's equation of order u is.

24 [Foundational] • Target: Complex roots

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

a. **5** ✓

b. -5

c. 6

d. -6

The given differential equation is $y'' - 5y' + 6y = 0$. The auxiliary equation for a second-order linear homogeneous differential equation of the form $ay'' + by' + cy = 0$ is $ar^2 + br + c = 0$. In this case, $a = 1$, $b = -5$, and $c = 6$. So the auxiliary equation is $r^2 - 5r + 6 = 0$. Let the roots of this quadratic equation be r_1 and r_2. According to Vieta's formulas, for a quadratic equation $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2$ is equal to $-\frac{b}{a}$. In our auxiliary equation.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Classifying without fully solving

Consider the second-order linear homogeneous differential equation $y''(x) + 5y'(x) + 6y(x) = 0$. If the auxiliary equation has roots r_1 and r_2, what is the value of $r_1 + r_2$?

a. 5

b. **-5** ✓

c. 6

d. -6

The general form of a second-order linear homogeneous differential equation with constant coefficients is $ay''(x) + by'(x) + cy(x) = 0$. The associated auxiliary equation is $ar^2 + br + c = 0$. For the given equation $y''(x) + 5y'(x) + 6y(x) = 0$, the auxiliary equation is $r^2 + 5r + 6 = 0$. According to Vieta's formulas, for a quadratic equation of the form $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2 = -b/a$. In this case, $a = 1$, $b = 5$, and $c = 6$. Therefore, the sum of the roots is.

26 [Foundational] • Target: Application: a damped spring

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the characteristic equation has roots r_1 and r_2 with $r_1 < r_2$, what is the value of r_1?

a. 1

b. **2** ✓

c. 3

d. 6

The characteristic equation for the given differential equation $y'' - 5y' + 6y = 0$ is found by replacing y'' with r^2, y' with r, and y with 1. This gives us the quadratic equation $r^2 - 5r + 6 = 0$. To find the roots, we can factor this quadratic equation: $(r - 2)(r - 3) = 0$. The roots are therefore $r = 2$ and $r = 3$. We are given that $r_1 < r_2$. Thus, $r_1 = 2$ and $r_2 = 3$. The question asks for the value of r_1, which is 2.

27 [Foundational] • Target: Verifying general solutions and the superposition principle

For which values of k does every nonzero solution of $y'' + 4y' + ky = 0$ genuinely oscillate (the auxiliary equation has non-real roots)?

a. $k < 4$

b. $k = 4$

c. $k > 0$

d. $k > 4$ ✓

The auxiliary equation $r^2 + 4r + k = 0$ has discriminant $16 - 4k$. Non-real roots require $16 - 4k4$ — exactly the range producing Theorem 3.2's complex-root case.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Boundary value problems

Consider the second-order linear homogeneous differential equation: $x^2y'' + xy' + (x^2 - u^2)y = 0$, where u is a constant. This is Bessel's equation of order u. Assume $u = 1/2$ and that $x > 0$. x is a regular singular point. Determine the indicial equation for this differential equation and find its roots.

a. **Indicial equation:** $r^2 - \frac{1}{4} = 0$**; roots:** $r = \pm\frac{1}{2}$ ✓

b. Indicial equation: $r^2 - r + \frac{1}{4} = 0$; roots: $r = \frac{1}{2}, \frac{1}{2}$

c. Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{4}$

d. Indicial equation: $r^2 + \frac{1}{4} = 0$; roots: $r = \pm\frac{i}{2}$

The given differential equation is $x^2y'' + xy' + (x^2 - u^2)y = 0$. We can rewrite this in the standard form $y'' + P(x)y' + Q(x)y = 0$ by dividing by x^2: $y'' + \frac{1}{x}y' + \frac{x^2 - u^2}{x^2}y = 0$. Thus, $P(x) = \frac{1}{x}$ and $Q(x) = \frac{x^2 - u^2}{x^2} = 1 - \frac{u^2}{x^2}$. $x = 0$ is a singular point. To determine if it's a regular singular point, we examine $xP(x)$ and $x^2Q(x)$. $xP(x) = x \cdot \frac{1}{x} = 1$, which is analytic at $x = 0$..

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The auxiliary equation

Consider the second-order linear homogeneous differential equation: $y'' - xy' + y = 0$. The point $x = 0$ is an ordinary point. Find the first two non-zero terms of the power series solution around $x = 0$, assuming $y(0) = 1$ and $y'(0) = 0$.

a. $1 + \frac{1}{2}x^2 + \ldots$ ✓

b. $1 + \frac{1}{3}x^3 + \ldots$

c. $1 - \frac{1}{2}x^2 + \ldots$

d. $1 - \frac{1}{3}x^3 + \ldots$

Let the power series solution be $y(x) = \sum_{n=0}^{\infty} a_n x^n$. Then $y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$. Substituting into the differential equation $y'' - xy' + y = 0$: $\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x\sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0$ $\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$ To combine the series, we shift the index in the first ter.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Distinct real roots

Consider the nonhomogeneous second-order linear differential equation $y'' - y = \frac{1}{1+e^x}$. If $y_1(x) = e^x$ and $y_2(x) = e^{-x}$ are two linearly independent solutions to the associated homogeneous equation, find the Wronskian of y_1 and y_2.

a. 2

b. **-2** ✓

c. 0

d. $e^x - e^{-x}$

The Wronskian of two functions $y_1(x)$ and $y_2(x)$ is defined as $W(y_1, y_2)(x) = \det(y_1(x), y_2(x)\ y_1'(x), y_2'(x))$. Given $y_1(x) = e^x$ and $y_2(x) = e^{-x}$, their derivatives are $y_1'(x) = e^x$ and $y_2'(x) = -e^{-x}$. Substituting these into the Wronskian formula: $W(y_1, y_2)(x) = \det(e^x, e^{-x}\ e^x,\ -e^{-x})\ W(y_1, y_2)(x) = (e^x)(-e^{-x}) - (e^{-x})(e^x)$ $W(y_1, y_2)(x) = -e^{x-x} - e^{-x+x}\ W(y_1, y_2)(x) = -e^0 - e^0\ W(y_1, y_2)(x) = -1 - 1$.

31 [Intermediate] • Target: Repeated roots

A damped spring satisfies $y'' + 2y' + 10y = 0$, $y(0) = 0$, $y'(0) = 6$. Find $y(\pi/6)$.

a. $2e^{\pi/6}$

b. $2e^{-\pi/6}$ ✓

c. $-2e^{-\pi/6}$

d. $e^{-\pi/6}$

Auxiliary: $r^2 + 2r + 10 = 0 \Rightarrow r = \frac{-2 \pm \sqrt{4-40}}{2} = -1 \pm 3i$ — $\alpha = -1, \beta = 3$, so $y = e^{-x}(C_1 \cos 3x + C_2 \sin 3x)$. $y(0) = C_1 = 0$, so $y = C_2 e^{-x} \sin 3x$; $y' = C_2 e^{-x}(3\cos 3x - \sin 3x)$, $y'(0) = 3C_2 = 6 \Rightarrow C_2 = 2$. So $y = 2e^{-x} \sin 3x$, and $y(\pi/6) = 2e^{-\pi/6} \sin(\pi/2) = 2e^{-\pi/6}$.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Complex roots

Consider the second-order linear non-homogeneous differential equation $y'' + 4y = f(t)$, where $f(t)$ is a piecewise constant function defined as $f(t) = \{2, , 0 \le t < 1; 0, , t \ge 1$. The Laplace transform of $f(t)$ can be written in the form $F(s) = \frac{2}{s} - \frac{2e^{-s}}{s}$. Find the Laplace transform of the differential equation, $s^2Y(s) - sy(0) - y'(0) + 4Y(s) = F(s)$, given initial conditions $y(0) = 1$ and $y'(0) = 0$. What is the expression for $Y(s)$?

a. $Y(s) = \frac{1}{s-1} + \frac{2}{s(s^2+4)} - \frac{2e^{-s}}{s(s^2+4)}$

b. $Y(s) = \frac{s}{s^2+4} + \frac{2}{s(s^2+4)} - \frac{2e^{-s}}{s(s^2+4)}$ ✓

c. $Y(s) = \frac{1}{s^2+4} + \frac{2}{s(s^2+4)} - \frac{2e^{-s}}{s(s^2+4)}$

d. $Y(s) = \frac{s}{s^2+4} + \frac{2}{s} - \frac{2e^{-s}}{s}$

The given differential equation is $y'' + 4y = f(t)$ with $f(t) = \{2, , 0 \le t < 1; 0, , t \ge 1$. The Laplace transform of $f(t)$ is $F(s) = \frac{2}{s} - \frac{2e^{-s}}{s}$. The initial conditions are $y(0) = 1$ and $y'(0) = 0$. The Laplace transform of the derivatives are: $\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$ and $\mathcal{L}\{y'(t)\} = sY(s) - y(0)$. Applying the Laplace transform to the differential equation: $s^2Y(s) - sy(0) - y'(0) + 4Y(s) = F(s)$. Substituting the initial condi.

33 [Intermediate] • Target: Classifying without fully solving

Consider the nonhomogeneous second-order linear differential equation $y'' + 4y = 8$. What is the particular solution, $y_p(x)$, to this equation?

a. 2 ✓

b. $2x$

c. $2\sin(2x)$

d. $2\cos(2x)$

The given differential equation is $y'' + 4y = 8$. This is a nonhomogeneous second-order linear differential equation. We use the method of undetermined coefficients to find a particular solution, $y_p(x)$. The nonhomogeneous term is a constant, 8. We guess a particular solution of the form $y_p(x) = A$, where A is a constant. Then, $y_p'(x) = 0$ and $y_p''(x) = 0$. Substituting these into the differential equation: $0 + 4(A) = 8$. Solving for A, we get $4A = 8$, which means $A = 2$. Therefore, the particular solut.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Application: a damped spring

Consider the nonhomogeneous linear second-order differential equation $y'' + y = 5$. If $y_p(x) = A$ is a particular solution, what is the value of A?

a. 1

b. **5** ✓

c. 1/5

d. 0

The given differential equation is $y'' + y = 5$. We are looking for a particular solution of the form $y_p(x) = A$, where A is a constant. To find the value of A, we substitute $y_p(x)$ and its derivatives into the differential equation. If $y_p(x) = A$, then the first derivative is $y_p'(x) = 0$ and the second derivative is $y_p''(x) = 0$. Substituting these into the differential equation $y'' + y = 5$, we get: $0 + A = 5$ Therefore, $A = 5$.

35 [Intermediate] • Target: Verifying general solutions and the superposition principle

Consider the second-order linear nonhomogeneous differential equation: $y''(x) - 3y'(x) + 2y(x) = 4$. Find the particular solution, $y_p(x)$, using the method of Undetermined Coefficients.

a. $y_p(x) = 2$ ✓

b. $y_p(x) = 2x$

c. $y_p(x) = 4$

d. $y_p(x) = 4x$

The given differential equation is $y''(x) - 3y'(x) + 2y(x) = 4$. Since the right-hand side is a constant, we guess a particular solution of the form $y_p(x) = A$, where A is a constant. Then, $y_p'(x) = 0$ and $y_p''(x) = 0$. Substituting these into the differential equation: $0 - 3(0) + 2(A) = 4$ $2A = 4$ $A = 2$ Therefore, the particular solution is $y_p(x) = 2$.

36 [Intermediate] • Target: Boundary value problems

Consider the second-order linear nonhomogeneous differential equation $y'' - y = \sin x$. If $y_p(x) = A\cos x + B\sin x$ is a particular solution, find the values of the constants A and B.

a. $A = 0, B = \frac{1}{2}$

b. $A = -\frac{1}{2}, B = 0$

c. $A = 0, B = -\frac{1}{2}$ ✓

d. $A = \frac{1}{2}, B = 0$

We are given the differential equation $y'' - y = \sin x$ and a proposed particular solution $y_p(x) = A\cos x + B\sin x$. We need to find the derivatives of $y_p(x)$ and substitute them into the differential equation. First derivative: $y_p'(x) = -A\sin x + B\cos x$. Second derivative: $y_p''(x) = -A\cos x - B\sin x$. Substitute these into the differential equation: $(-A\cos x - B\sin x) - (A\cos x + B\sin x) = \sin x$ Combine like terms: $(-A - A)\cos x + (-B - B)\sin x = \sin x$.

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: The auxiliary equation

Find $y(2)$ for the differential equation $y' - y = 4$ satisfying $y(0) = 1$.

a. $5e^2 + 4$

b. $5e^2 - 4$ ✓

c. $e^2 - 4$

d. $4e^2 - 5$

$\mu = e^{\int -1\,dx} = e^{-x}$. Multiplying gives $(ye^{-x})' = 4e^{-x}$. Integrating gives $ye^{-x} = -4e^{-x} + C \Rightarrow y(x) = -4 + Ce^{x}$. Applying $y(0) = 1$ yields $1 = -4 + C \Rightarrow C = 5$. Thus $y(x) = 5e^{x} - 4$, so $y(2) = 5e^2 - 4$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Distinct real roots • Cross-Lesson: 5.01 Variation of Parameters

An RC circuit with $R = 20\,\Omega$ and $C = 0.01$ F is driven by $E(t) = 100\sin(10t)$ V. What is the first-order differential equation for the charge $q(t)$?

a. $q' + 10q = 5\sin(10t)$

b. $q' + 5q = 10\sin(10t)$

c. $q' + 5q = 5\sin(10t)$ ✓

d. $20q' + 5q = \sin(10t)$

The circuit law gives $Rq' + \frac{1}{C}q = E(t) \Rightarrow 20q' + 100q = 100\sin(10t)$. Dividing by 20 gives $q' + 5q = 5\sin(10t)$.

39 [Challenging] • Target: Repeated roots • Cross-Lesson: 1.01 Separable Equations

Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:

a. $e^{i\beta x}$

b. $C_1 e^{\alpha x} + C_2 e^{\beta x}$

c. $(C_1 + C_2 x)e^{(\alpha+\beta)x}$

d. $e^{\alpha x}(C_1 \cos\beta x + C_2 \sin\beta x)$ ✓

Euler: $e^{\alpha x}$ envelope times sinusoids.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Complex roots • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the differential equation $\frac{dy}{dx} = \frac{\cos(x)}{y^2}$ for $y > 0$ and $x \in [0, \frac{\pi}{2}]$. If $y(0) = 2$, find the value of $y(\frac{\pi}{2})$.

a. $\sqrt[3]{6}$ ✓

b. $\sqrt[3]{3}$

c. $\sqrt[3]{9}$

d. $\sqrt[3]{12}$

The given differential equation is $\frac{dy}{dx} = \frac{\cos(x)}{y^2}$. This is a separable differential equation. We can rewrite it as $y^2 dy = \cos(x) dx$. Integrating both sides, we get: $\int y^2 dy = \int \cos(x) dx$ $\frac{y^3}{3} = \sin(x) + C$ We are given the initial condition $y(0) = 2$. Substituting these values into the integrated equation: $\frac{2^3}{3} = \sin(0) + C$ $\frac{8}{3} = 0 + C$ $C = \frac{8}{3}$ So, the particular solution is: $\frac{y^3}{3} = \sin(x) + \frac{8}{3}$ We need to find the.

41 [Challenging] • Target: Classifying without fully solving • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Consider the system of linear differential equations given by: $\begin{cases} \frac{dx}{dt} = -x + 2y; \frac{dy}{dt} = 3x - 2y \end{cases}$ Let the general solution be of the form $\mathbf{x}(t) = c_1\mathbf{v}_1 e^{\lambda_1 t} + c_2\mathbf{v}_2 e^{\lambda_2 t}$, where $\lambda_1 < \lambda_2$. Determine the value of λ_2.

a. 1

b. 2

c. **3** ✓

d. 4

The given system can be written in matrix form as $\frac{d\mathbf{x}}{dt} = A\mathbf{x}$, where $A = (-1, 2; 3, -2)$. To find the eigenvalues λ, we solve the characteristic equation $\det(A - \lambda I) = 0$. $\det(-1 - \lambda, 2; 3, -2 - \lambda) = 0$ $(-1 - \lambda)(-2 - \lambda) - (2)(3) = 0$ $(1 + \lambda)(2 + \lambda) - 6 = 0$ $2 + \lambda + 2\lambda + \lambda^2 - 6 = 0$ $\lambda^2 + 3\lambda - 4 = 0$ Factoring the quadratic equation: $(\lambda + 4)(\lambda - 1) = 0$ The eigenvalues are.

42 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 5.01 Variation of Parameters

Consider the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$. If the solution curve passes through the point $(1, 2)$, find the value of the constant of integration, C, when the equation is integrated in the form $y^2 = x^2 + C$.

a. 1

b. 2

c. **3** ✓

d. 4

The given differential equation is $\frac{dy}{dx} = \frac{x}{y}$. This is a separable equation. We can rewrite it as $y\,dy = x\,dx$. Integrating both sides, we get: $\int y\,dy = \int x\,dx$ This yields $\frac{1}{2}y^2 = \frac{1}{2}x^2 + C_1$. Multiplying by 2, we get $y^2 = x^2 + 2C_1$. Let $C = 2C_1$. Thus, the integrated form is $y^2 = x^2 + C$. We are given that the solution curve passes through the point $(1, 2)$. We can substitute these values into the integrated equation to find the value of C.

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Verifying general solutions and the superposition principle • Cross-Lesson: 1.01 Separable Equations

A tank holds 60 L of pure water. Brine with concentration 2 kg/L enters at 4 L/min, and the thoroughly mixed solution leaves at 2 L/min. What is the salt concentration in the tank after 30 minutes?

a. 1.5 kg/L ✓

b. 1.2 kg/L

c. 1.8 kg/L

d. 2.0 kg/L

$V(t) = 60 + 2t$. Salt equation: $S' + \frac{1}{t+30}S = 8$. Integrating factor: $\mu = t + 30$. Multiplying gives $((t+30)S)' = 8(t+30)$. Integrating yields $(t+30)S = 4(t+30)^2 + C$. Using $S(0) = 0$ yields $C = -3600$. At $t = 30$, $t + 30 = 60$, so $S(30) = 4(60) - \frac{3600}{60} = 240 - 60 = 180$ kg. The volume is $V(30) = 120$ L, so the concentration is $c = \frac{180}{120} = 1.5$ kg/L.

44 [Challenging] • Target: Boundary value problems • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y' + y\sec x = \cos x$ on $(-\pi/2, \pi/2)$ with $y(0) = 0$. What is the value of $y(\pi/3)$?

a. $\frac{2\pi+3}{6}$

b. $\frac{(2\pi+3)(2+\sqrt{3})}{6}$

c. $\frac{(2\pi+3)(2-\sqrt{3})}{6}$ ✓

d. $\frac{(2\pi-3)(2-\sqrt{3})}{6}$

$\mu = e^{\int \sec x\,dx} = \sec x + \tan x$. Multiplying gives $(y(\sec x + \tan x))' = \cos x(\sec x + \tan x) = 1 + \sin x$. Integrating yields $y(\sec x + \tan x) = x - \cos x + C$. Using $y(0) = 0$ gives $0 = 0 - 1 + C \Rightarrow C = 1$. At $x = \pi/3$, $\sec(\pi/3) + \tan(\pi/3) = 2 + \sqrt{3}$, and the right side is $\pi/3 - 1/2 + 1 = \frac{2\pi+3}{6}$. Dividing gives $y(\pi/3) = \frac{(2\pi+3)(2-\sqrt{3})}{6}$.

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: The auxiliary equation • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A tank contains 50 gal of pure water. Brine with 2 lb/gal of salt enters at 3 gal/min, and the well-stirred solution drains out at 1 gal/min. How much salt is in the tank when the volume reaches 100 gallons?

a. $200 - 25\sqrt{2}$ lb

b. 150 lb

c. $200 - 50\sqrt{2}$ lb ✓

d. $200 - 50\sqrt{3}$ lb

$V(t) = 50 + 2t = 100 \Rightarrow t = 25$ min. Salt equation: $S' + \frac{1}{2(t+25)}S = 6$. Integrating factor: $\mu = \sqrt{t+25}$. Multiplying gives $(\sqrt{t+25}\,S)' = 6\sqrt{t+25}$. Integrating yields $\sqrt{t+25}\,S = 4(t+25)^{3/2} + C$. Using $S(0) = 0$ gives $0 = 4(125) + C \Rightarrow C = -500$. At $t = 25$, $t + 25 = 50$, so $S(25) = 4(50) - \frac{500}{\sqrt{50}} = 200 - 50\sqrt{2}$ lb.

© ClipSAT — clipsat.org | All Rights Reserved